

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: 1.7 MICROORGANISMS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Remember the sufuria of leftover ugali and the half-eaten chapati left uncovered on a kitchen counter in Kisumu overnight. By morning there were white, green and black fuzzy patches, a sour smell and some slimy areas. Nobody had touched the food, no insects were visible and the kitchen looked clean. Use what you learned in Lessons 1 to 8 to answer the Driving Question: Why did the ugali and chapati grow fuzzy patches overnight, and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya? Write your answers in the space under each prompt. Use full sentences, name the microorganisms and conditions you studied, and refer to the Kisumu ugali and chapati in every answer. Where a prompt asks for evidence, say what you observed or learned in class.

Part 1: What are the patches and where did they come from?

Look back at the photographs of the Kisumu ugali and chapati. (a) Are the white, green and black patches living or non-living? What type of microorganism caused them? (b) Nobody touched the food and the kitchen looked clean. Explain where the organisms came from and how they reached the food. (c) The patches were different colours. What does this tell you about the organisms that grew on the food?

Write your answer here:

Part 2: Why did it happen overnight?

In Lesson 7 you compared mould growth on ugali, bread and dry maize biscuit under different conditions (covered or uncovered, moist or dry, warm or cool). (a) Which three conditions favoured mould growth, and what did the pre-incubated samples show about them? (b) Use these conditions to explain why the Kisumu ugali and chapati spoiled so quickly. (c) Why is the dry maize biscuit useful as a comparison?

Write your answer here:

Part 3: Protecting our food and our health

(a) Choose TWO ways of protecting the ugali and chapati from spoilage. For each, say which condition or transmission route it interrupts. (b) Some Kisumu households, such as Mama Adhiambo's in Nyalenda, may not have a refrigerator. Suggest low-cost options and explain why they work. (c) Name TWO traditional Kenyan food preservation methods from our lessons and explain why they work.

Write your answer here:

Part 4: The wider world of microorganisms

(a) The mould on the ugali belongs to the fungi. Name one infection caused by fungi, one by bacteria and one by viruses that you studied, and say how each is spread. **(b)** Why does knowing the mode of transmission help us choose the best way to prevent an infection? **(c)** Microorganisms are not always harmful. Give TWO beneficial roles from Lesson 6 and explain how the same group of organisms can be both a problem and useful.

Write your answer here:

Part 5: Answer the Driving Question

Write a short message of about 5 to 6 sentences for households in your own locality. In it, explain why the ugali and chapati grew fuzzy patches overnight and give TWO pieces of advice to protect food and health. Include at least one piece of evidence from our lessons.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explanation of the phenomenon (what and where from)	Clearly identifies the patches as living fungal mould colonies, explains airborne spores as the source and route, and explains the colour variation as different species.	Identifies the patches as fungal mould and says spores came from the air, but gives limited explanation of the colour variation.	Says the patches are germs or dirt, or does not explain how the organisms reached the food.
Use of evidence on growth conditions	Names warmth, moisture and air exposure, uses the Lesson 7 observations, and links all three to the overnight spoilage in Kisumu.	Names the conditions and links most of them to the phenomenon, with limited use of investigation evidence.	Names one condition or gives a vague link to the phenomenon, with no evidence.
Prevention and control reasoning	Matches each control measure to the route or condition it interrupts and suggests accessible, low-cost options for Kenyan households.	Gives suitable control measures with some explanation of how they work.	Lists measures without explaining why they work.
Scientific knowledge of	Accurately describes infections,	Mostly accurate examples with minor	Few or inaccurate examples.

microorganisms	transmission routes and beneficial and harmful roles, and shows how the same group can be both.	gaps in transmission or the dual role.	
Communication and Kenyan context	Clear, well-organised message for the community that uses accurate scientific terms and local examples.	Clear message with some scientific terms and some local context.	Message is unclear or lacks scientific terms and local context.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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